BERK DEMIRCI

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SUMMARY

Final-year Computer Engineering student at TED University, part-time AI engineer at Enocta, and independent researcher at Eschatia Labs; focused on Machine Learning, Natural Language Processing (NLP), and AI Agents / Fine-tuning.

EDUCATION

TED University, Ankara, Turkiye
Computer Engineering, GPA: 3.70, 4th Year (Senior)

ASO Technical College High School, Ankara, Turkiye
Department: Industrial Automation Technologies / Mechatronics

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL (MySQL), Java, PHP, JavaScript, C
- **Tools:** Excel, Power BI, Git, OpenAI API, LangChain, LangGraph, AI Agents, RAG Pipelines, Hugging Face, MCP (Model Context Protocol)
- **Python Libraries:** Machine Learning and Automation Libraries (PyTorch, scikit-learn, pandas, numpy, etc.)
- **Languages:** Advanced English

EXPERIENCE

Enocta – AI Engineer (Part-time), August 2026 – Present
- Fine-tuning and agent harness development for local language models; actively using enterprise-grade AI tools such as ElevenLabs.
- Joined as an AI intern (August 2026 – September 2026) and continued as a part-time AI engineer.

Eschatia Labs – Independent Researcher, Present
- International open-source research lab (eschatialabs.com) building AI models, datasets, and papers: deepfake detection, multilingual decision models, local AI assistants, and language preservation with small models.

OBSS, July 2026 – August 2026
- Developed AI-powered applications as an AI intern.
- Placed first in an internship-program competition.

Aselsan, June 2026 – July 2026
- Tested the performance of a local AI model and improved it using the collected data.

Pulsar Robotik, July 2025 – August 2025
- AI-supported automation with robot arms (visual-language-action)
- Chemical dosing machine automations
- Computer vision applications with SensoPart cameras for quality control

TNC, April 2025 – May 2025
- Developed projects using MS Office, Photoshop, and AutoCAD

Ankara Klima, July 2022 – August 2022
- MCC/DDC (Automation Panel) preparation
- Air Handling Unit analysis

ASO Technical High School, June 2021 – July 2021
- Sensor Data Analysis and Visualization – real-time data collected from Arduino sensors (temperature, motion), trend analysis in Python
- STM32 microcontroller programming
- Robotic hand control and automation

PROJECTS

- **koku** – Turkish perfume search: an 8B teacher model writes synthetic data, a 0.6B student is distilled from it, and a fine-tuned retriever lifts nDCG@10 from 0.18 to 0.48 on 183 Turkish queries. (Distillation, Retrieval, Fine-tuning)

▪ **Fly Wardrobe** – Interactive 3D demo where a fruit fly picks a perfume for six destinations using the koku retriever, then tracks the odor plume while its FlyWire connectome brain lights up. (WebGL, FlyWire, Retrieval)
▪ **Personal AI Assistant** – Built a personal AI assistant that runs deep web research (Parallel.AI), social research on Reddit and X, keeps second-brain notes, and turns results into personalized PDF roadmaps; started as an L&D assistant and grew into a general assistant. (Python, React, Hermes Agent)
▪ **Open-Ended Questions** – App that generates 4–6 open-ended interview questions and a shared 1–4 scoring rubric from a job description, scores answers with TypeSafe Jev, flags low-confidence scores (below 60%) for human review, and rejects off-topic and prompt-injection inputs. (Next.js, OpenRouter)
▪ **Jev ATS** – Test bench for rubric-based CV screening with TypeSafe Jev across 6 role rubrics; repeats each evaluation to measure score consistency. (FastAPI)
▪ **CV Sender** – LLM-powered software that analyzes user resumes and company goals to identify the best matches from a pool and auto-generates tailored emails. (Web Scraping, AI)
▪ **UniBot** – RAG/LLM-powered software that scrapes a university website, creates a vector database, and answers user questions using BM25, RAG, and a reranker. (Web Scraping, AI)
▪ **Medical Expense Prediction** – End-to-end machine learning project predicting medical service costs based on patient data (deployed to production, self-retraining system). (Machine Learning-MLOps)
▪ **Air Quality Index Prediction** – Model predicting air quality from weather and pollutant measurements. (Machine Learning, Time Series)
▪ **Pokemon Card Game Database** – Pokemon card game web app using SQL and Blazor. (Database)
▪ **University Course Scheduler** – System generating course schedules based on student, professor, and time constraints (SQL and Python). (Data Analysis)
▪ **Wikipedia Biography Summarizer** – Python-based voice assistant using web scraping to summarize biographies on Wikipedia. (Data Analysis & Automation)
▪ **DunGen AI** – A React Native-based mobile app that supports Dungeons & Dragons-style tabletop RPGs, offering AI-powered content generation and features that enrich the gameplay experience; currently in closed beta on Google Play with test users. (Mobile App, AI)

OPEN SOURCE CONTRIBUTIONS

▪ **Natively Cluely AI Assistant** - Enhanced the repository's robustness by developing a token fallback method for OpenAI API calls, automatically handling parameter compatibility.
▪ **World Monitor** - Contributed a pull request adding a collapsible emoji legend to the Strategic Posture panel, clarifying the meaning of military-asset emojis to reduce user confusion.
▪ **GitNexus** - Contributed a pull request fixing an 'undefined' parsing error caused by missing language mappings (e.g. Kotlin) in the CLI and Web call routers, and added a generic fallback handler for additional robustness.

SOCIETIES

▪ Google Developer Groups On Campus TEDU – President
▪ IAESTE TEDU – Board Member
▪ Conducted AI/ML hands-on workshops for students with Colab Notebooks and Python
▪ Organized a hackathon for an LLM-powered website

COURSES & CERTIFICATIONS

▪ Machine Learning Specialization – Andrew Ng, Stanford University, DeepLearning.AI
▪ IBM Data Science – IBM
▪ Computer Science: Goal-Oriented Programming – Princeton University

COMPETITIONS

▪ Insider's Datathon, BTK Datathon, ING Datathon – machine learning predictions to improve customer satisfaction using real-world datasets

PERSONAL SKILLS

Problem Solving, Critical Thinking, Analytical Thinking, Leadership, Teamwork, Communication, Punctuality

HOBBIES

Reading (Novels, AI/ML Research papers), Swimming, Fitness, Photography